Grade 3 Coding: Hello Python

Lesson Guide & Handout — Writing Your First Text Program!

1. Introduction to Python

Python is a powerful programming language. It tells the computer exactly what to do, step by step! Professional developers use Python all over the world to make amazing games, mobile apps, and websites.

Fun fact: Python wasn't named after a snake! It was actually named after a famous comedy television show.

2. Meeting the Code Editor

To give instructions to Python, we type our code inside a software called an **editor**. Popular options include *Thonny*, *IDLE*, or various student-friendly online editors. Always confirm with your teacher which one to launch in class.

A standard Python editor is split into two major spaces:

- **Code Area:** The main workspace where you compose and type your instructions.
- **Output Area:** The console box where results instantly appear after you execute your project.

Whenever you finish writing your commands, click the **Run** button to see the magic happen!

3. Your Very First Command: `print()`

The `print()` command tells Python to display text onto your screen. Whatever valid content you supply inside the brackets will be shown directly in the output area.

Let's Try It Together

Type the following exact command inside your editor's code area:

```
print("Hello, World!")
```

Press the **Run** button. The computer will output:

```
Hello, World!
```

4. Understanding Strings and Quotes

In coding, plain text is referred to as a **string**. Strings represent regular sentences, words, or collections of characters.

- `"Hello"` is a string.
- `"Python is fun"` is also a string.
- Strings must always be wrapped completely inside single quotes `' '` or double quotes `" "`.

⚠️ **Avoid Common Mistakes!**

Without proper matching quotes, Python gets completely confused and throws a program error.

Right: `print("Hi there!")`

Wrong: `print(Hi there!)` ← *This will cause an Error!*

Wrong: `print "Hello"` ← *Forgetting brackets entirely causes an Error!*

Always remember the standard pattern: `print("your text here")`

5. Printing Multiple Lines

Every single `print()` command you execute automatically generates a completely new line in the console. Try typing these three instructions sequentially:

```
print("My name is Python")
print("I am learning to code")
print("This is so cool!")
```

Your output will cleanly separate across three distinct lines:

```
My name is Python
I am learning to code
This is so cool!
```

6. Working with Variables

Think of a **variable** as a labeled storage box that holds information inside the computer's memory. You can name these boxes and give them values using the equals sign (=).

```
name = "Aarav"
age = 8
```

Now, the variable `name` holds the string value `"Aarav"`, and `age` holds the number `8`!

Printing a Variable

You can print the contents of a variable box by placing the variable name inside the brackets. **Crucial rule: Do not place quotes around your variable name when printing it!**

```
name = "Priya"
print(name)
```

The variable provides its stored value directly to the `print()` statement, and the console shows:

```
Priya
```

🚀 Mini Coding Challenge!

Open up your editor and create a brand-new program. Use three separate `print()` statements to print out your favorite things:

- Line 1: Your name
- Line 2: Your favorite food
- Line 3: Your favorite color

Example format to guide you: `print("My favorite color is Blue")`

7. Summary Checklist

Let's review everything we successfully mastered today:

✓ The `print()` command shows custom text results onto the user's screen.

✓ Strings are text collections and must always reside safely inside matching quotes.

✓ Variables act like labeled storage boxes to remember clean information for later.

✓ We can print the contents of variables directly without using any quotes around them.

✓ Always double-check your code punctuation for missing brackets or loose quotes!

Next Class: Input Operations — Learning how to ask the user questions and read their answers! See you soon, coder!